Machine Learning in Production

Lecture 2: *Requirements, Value, and Risk*

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Last class

- Machine learning components are part of larger systems
- Data scientists and software engineers have different goals and focuses
  - Building systems requires both
  - Various qualities are relevant, beyond just accuracy
- Machine learning brings new challenges and intensifies old ones

Meet UniGuide

- Your team has built **UniGuide**, an AI assistant for UET
- A student can ask:
  - “Can I graduate next semester?”
  - “Which course satisfies this prerequisite?”
  - “What happens if I withdraw from this course?”
  - “When is the scholarship application due?”





UniGuide, one month later…

- Offline acceptable-answer rate: 84% → 92%
- Routine questions answered faster: ↑
- Adviser workload: ↑
- Severe complaints: ↑
- Two students delayed graduation plans after following incomplete advice.



Did the system improve?

Your hypothesis for this issue?

The benchmark improved because ___, while the product outcome worsened because ___.

Should UniGuide use ML at all?



When ML is attractive

01 Language variation

Questions are varied and language-rich.

02 Change

Policies change several times each year.

03 Synthesis

Some answers combine faculty and university rules.

04 Scale

Demand spikes sharply during registration.

05 Coverage

Staff cannot author every possible wording.

06 Mixed workload

Most cases are routine; a minority are highly contextual.

Does this support ML, automation, or neither?

When ML adds risk

Irreversibility

A wrong scholarship deadline can create irreversible loss.

Rare severity

A small number of cases carry much higher consequences.

Simple baseline

Authoritative search is sufficient for many questions.

Weak source

The policy repository is incomplete and inconsistently structured.

Limited fallback

Adviser capacity is limited, but not zero.

Uneven coverage

Students use both English and Vietnamese.

Change your mind?

ML must earn its place!

ML’s place

Use ML when…

- The task is difficult to specify directly
- Scale, variation, or change defeat maintainable rules
- Relevant data and evaluation are feasible
- The workflow can detect or contain mistakes
- Value exceeds build, review, inference, and operating cost

Prefer simpler methods…

- Search or rules provide adequate guarantees
- Consequences demand strong predictability
- Evidence is too weak for the proposed scope
- Fallback capacity is unavailable
- The operating cost outweighs the expected benefit

A strong design may use ML for one bounded part
and preserve rules, search, human judgment, or no automation elsewhere.

Layers of Success Measures

- **Organizational objectives:** Innate/overall goals of the organization
- **Leading indicators:** Short-term proxies for long term measures, typically for organizational objectives
- **System goals:** Goals of the software system/product/feature to be built
- **User outcomes:** How well the system is serving its users, from the user's perspective
- **Model properties:** Quality of the model used in a system, from the model's perspective



The UniGuide goal chain?





Layers of Success Measures

At every link

ASSUMPTION

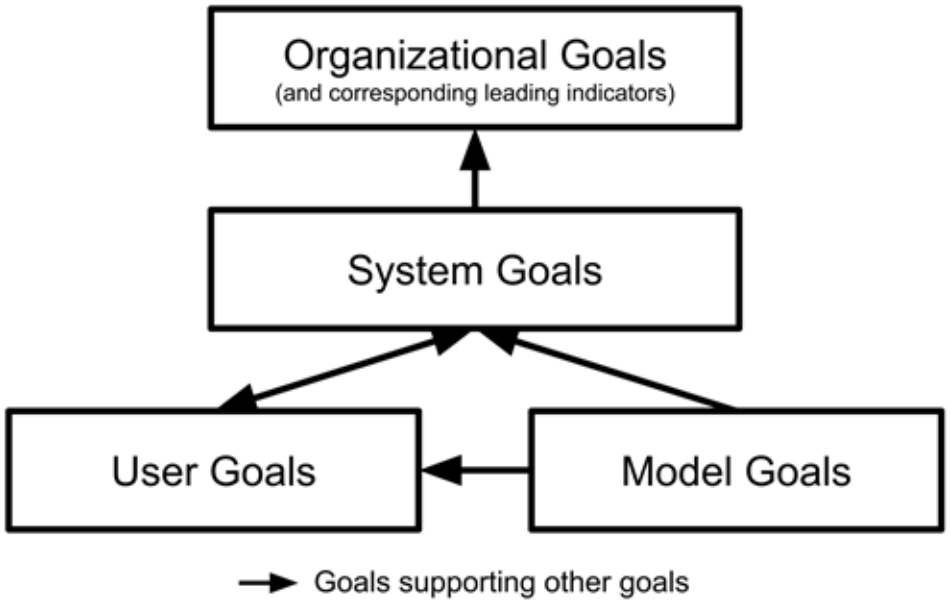
What must be true?

EVIDENCE

What would support or weaken it?

STAKEHOLDER

Who may experience a different outcome?



Proxy trap

Easy to measure	How the outcome may worsen
Session length ↑	Students remain because they are confused
Answer acceptance ↑	Fluent errors appear trustworthy
Adviser visits ↓	Escalation becomes difficult
Resolved chats ↑	Cases close before the real issue is solved
Average quality ↑	Severe failures remain hidden in a small group

Useful proxy = causal argument + balancing measure + test for gaming

Which success should we optimize?



Justify your choice:
priority + benefit + dangerous side effect + required guardrail.

Optimize a portfolio, not one number!

A defensible success portfolio

PRIMARY OUTCOME

User or product effect being pursued

GUARDRAILS

Harms or quality losses that must remain bounded

BALANCING MEASURES

Costs shifted to other stakeholders or workflows

LEADING INDICATORS

Signals available before the final outcome

MODEL DIAGNOSTICS

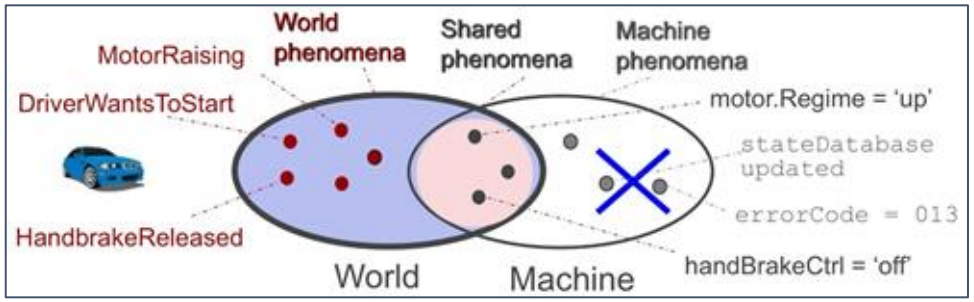
Component evidence - not the final goal

DECISION OWNER

Role accountable for the trade-off

World, machine, and the statements between them

- **Requirements (REQ)** are expressed only in terms of world phenomena
- **Assumptions (ASM)** expressed in terms of world & shared phenomena
- **Specifications (SPEC)** are expressed in terms of shared phenomena
- **Key point: Software cannot directly satisfy a requirement on its own; it relies on assumptions about the environment!**



Important to make assumptions explicit

- **System Requirement (REQ)**: What the system must ensure, in terms of desired effects on the *environment*
- **Software Specification (SPEC)**: What *software must implement*, expressed over the shared phenomena (input/output devices)
- **Assumptions (ASM)**: What's assumed about the behavior/properties of the environment; bridges the gap between REQ and SPEC

$$ASM \wedge SPEC \Rightarrow REQ$$

REQ, ASM, SPEC or Implementation

1	Students retain access to qualified human advice when the assistant abstains.	?
2	The authoritative policy repository is updated before a new rule takes effect.	?
3	If policy sources conflict, the assistant does not state an eligibility conclusion.	?
4	The deployed retriever ranks a superseded policy first.	?
5	Students interpret a confident answer as official approval.	?

Lane Keeping Assist

- **Requirement (REQ)**: The vehicle must be prevented from veering off the lane.
- **Specification (SPEC)**: ??



Lane Keeping Assist

- **Requirement (REQ)**: The vehicle must be prevented from veering off the lane.
- **Specification (SPEC)**: Lane detector accurately identifies lane markings in the input image; the controller generates correct steering commands



What assumptions (ASM) are needed?

What could go wrong?

Recall: ASM ∧ SPEC => REQ

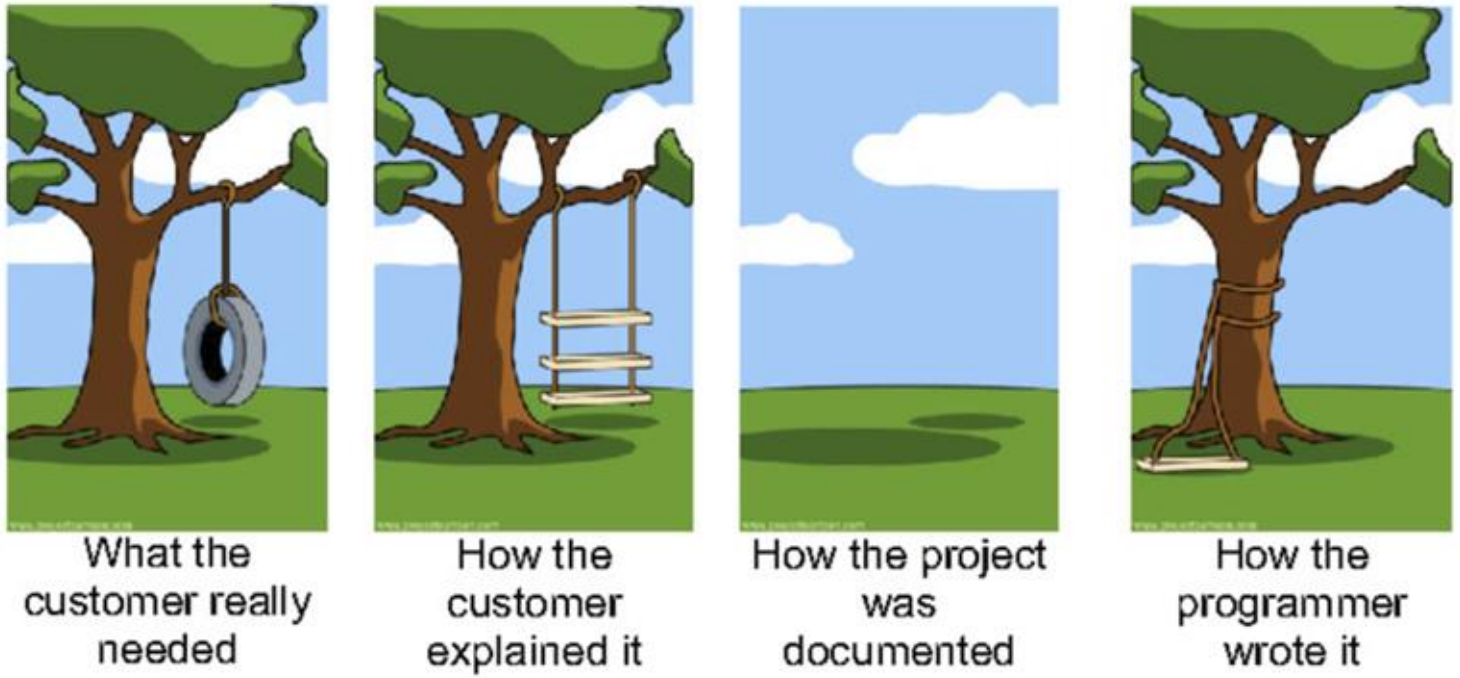
1. Wrong, inconsistent or infeasible requirements (REQ)
2. Missing or incorrect environmental assumptions (ASM)
3. Wrong or violated specification (SPEC)

Q. Example each for lane assist?

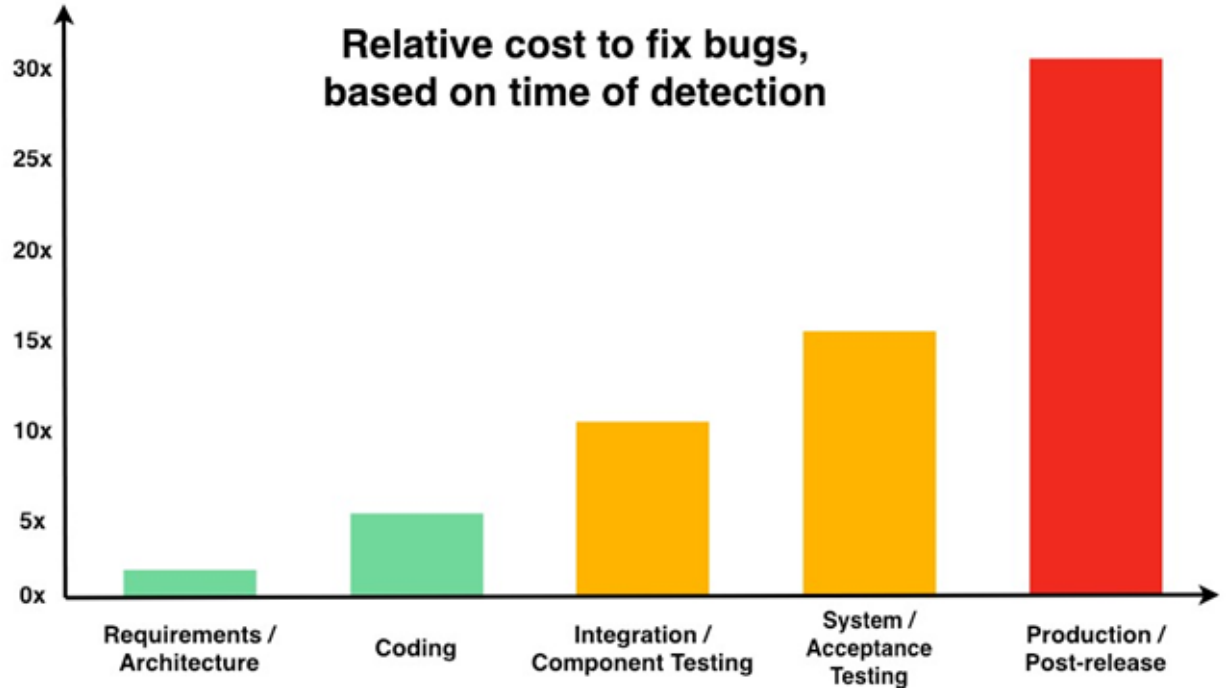


Software Requirements

Describe what the system should do, in terms of the services that it provides and their qualities (safety, reliability, performance...)



Facing costly problems later...



If you build the wrong system, it will be really expensive to fix it later!

Requirements & Design: **Think before coding**

Facial Recognition in ATM



Lufthansa 2904 Runaway Crash



Lufthansa 2904 Runaway Crash

Reverse thrust: Decelerates plane after landing

REQ: Reverse thrust is enabled if and only if plane is on the ground

SPEC: Reverse thrust is enabled if and only if wheel is turning

- *if (a) 6.3 tons of weight are sensed on each landing gear or (b) sensors indicate the wheels are turning faster than 72 knots*

Assumption: Wheel is turning if and only if plane on the ground

Assumption: High amounts of weight are only on both landing gears if the plane is on the ground

Lufthansa 2904 Runaway Crash



Lufthansa 2904 Runaway Crash

REQ: Reverse thrust is enabled if and only if plane is on the ground

SPEC: Reverse thrust is enabled if and only if wheel is turning

ASM: Wheel is turning if and only if plane on the ground

On that day, runway was wet due to rain!

- Wheel fails to turn, even though the plane is on the ground (assumption violated)
- Pilot attempts to enable RT; overridden by the software
- Plane goes off the runway and crashes!

Weak versus testable requirement

“UniGuide answers should be safe and accurate.”

What are still unresolved?

Weak versus testable requirement

“UniGuide answers should be safe and accurate.”

For deadline-sensitive eligibility questions using current authoritative policy corpus:

- no eligibility claim without a current citation
- conflict or insufficiency triggers escalation
- documented expert-review protocol
- separate reporting for programme and language groups

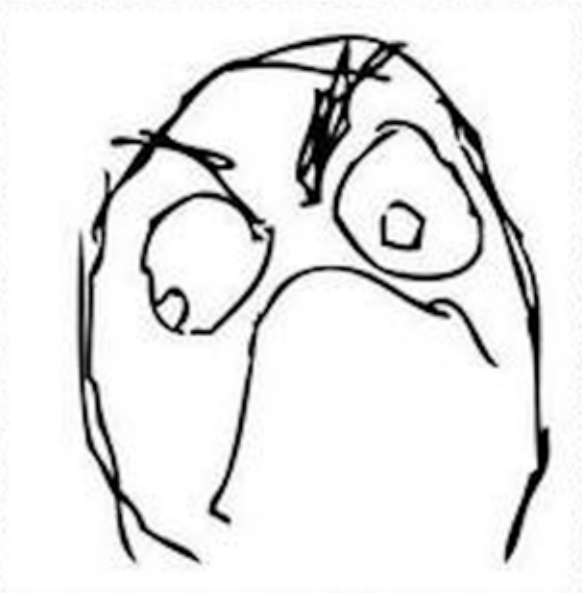
Understanding requirements is hard

Why?



“Yes…but” problem





Is 95% good enough?

The system answers 95% of evaluation questions acceptably.

A **YES**

Withholding a useful service also has a cost.

B **NO**

Five percent error is unacceptable for academic advice.

C **NOT ENOUGH**

The aggregate cannot answer the release question.

“Good enough” is a risk decision

A threshold becomes defensible only when tied to…

USE + SCOPE

intended use + bounded release

CONTEXT

population + task + conditions

CONSEQUENCE

error types + affected stakeholders

EVIDENCE

baseline + uncertainty + slices

BEHAVIOR

abstain + fallback + escalate

GOVERNANCE

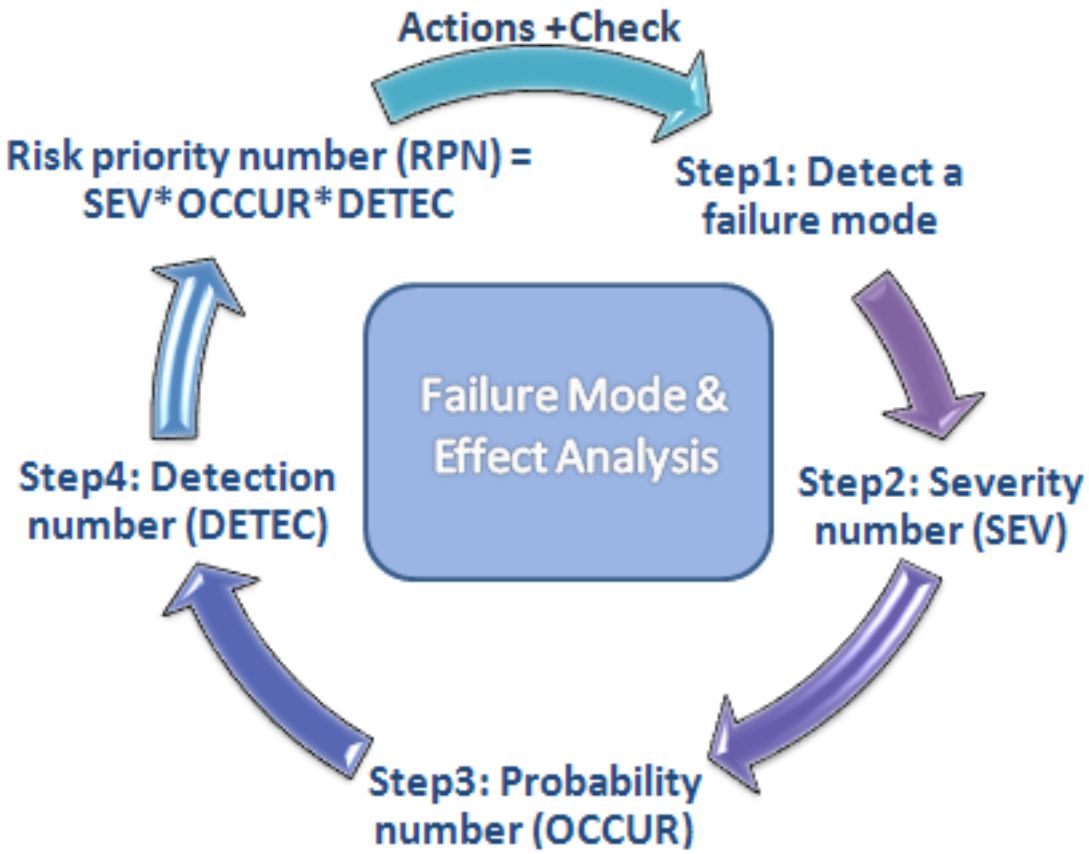
detect + recover + stop + owner

Risk as structured anticipation

- Risk statement:
  - Because **[cause or violated assumption]**, **[component/process]** may **[failure mode]**, causing **[system/user effect]** for **[stakeholder]**.
  - severity;
  - likelihood or exposure;
  - detectability;
  - reversibility and recovery difficulty.
- Pipeline:

  prevention → detection → response → recovery → owner → evidence

Failure mode and effects analysis



Exercise: Requirement and risk

For UniGuide system (or your selected one), provide:

- one bounded purpose and system scope;
- two measurable requirements;
- two critical environmental or organizational assumptions;
- three failures in 3 following failure mode:
  - Model-behavior failure
  - Data or environmental-assumption failure
  - Workflow, interface, or human-review failure
- one prevention -> detection -> response -> recovery chain;
- one owner and one unresolved evidence need.

Uncertainty is not an excuse

We cannot foresee every failure. We can still:

- make assumptions visible;
- seek perspectives from affected and operating roles;
- collect evidence for the weakest links;
- design detection and containment;
- provide recovery and recourse;
- assign authority and review triggers;
- record residual risk honestly.

Bounded purpose and system scope

- In scope:
  - Retrieve current, approved academic policies.
  - Compare a student's recorded courses with standard program requirements.
  - Explain the result and cite the relevant policy.
  - Escalate ambiguous cases to a human adviser.
- Out of scope:
  - Making the official graduation decision.
  - Approving substitutions, waivers, or policy exceptions.
  - Interpreting unpublished or future policy changes.
  - Advising postgraduate or non-degree students.

Two measurable requirements

- REQ-01 - Correct, grounded guidance
  - For in-scope undergraduate graduation questions, at least 95% of answers on a versioned, stratified evaluation set shall provide the correct eligibility guidance and cite the applicable current policy; performance shall not fall below 90% for any monitored program or entry-year slice.
- REQ-02 - Safe handling of uncertainty
  - When required records are missing, policies conflict, or an exception is needed, the system shall abstain and create an adviser-review case in at least 99% of such cases, rather than provide a definitive eligibility judgment. The case shall include the student’s question, relevant record fields, retrieved policy passages, and the reason for escalation.

Two critical assumptions

- ASM-01 - Policy freshness
  - The registrar publishes every approved policy change in an authoritative digital source with an effective date before the policy is applied to students.
- ASM-02 - Human-service capacity
  - Academic advising has sufficient staffing and authority to review escalated graduation cases within the promised service window, including during peak registration and graduation periods.

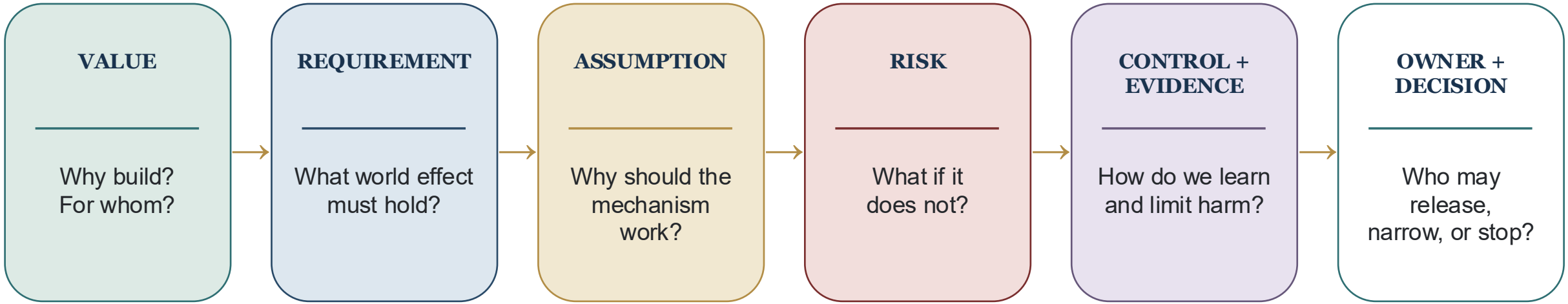
Three failure modes

Failure category	Example failure	Potential consequence
Model-behavior failure	The model treats an exception described in a retrieved document as a general rule and tells an ineligible student that they have satisfied the requirement.	False reassurance may delay corrective action and jeopardize graduation.
Data or environmental-assumption failure	The retrieval index contains last year's policy because a newly approved requirement was not published or ingested on time.	The system gives an answer that is grounded in a real but obsolete policy.
Workflow, interface, or human-review failure	The system escalates an ambiguous case, but the review ticket omits the graduation deadline and is not prioritized by the adviser queue.	A technically correct escalation still arrives too late to prevent harm.

Owner and unresolved evidence need

- Accountable owner:
  - UniGuide Service Owner: accountable for verifying that the complete prevention-detection-response-recovery chain operates across the registrar, engineering, and advising teams.
- Unresolved evidence need:
  - We do not yet know whether the advising service can process the projected number of escalations within the promised response time during peak periods. This must be measured through workload data or a controlled pilot before confirming ASM-02 or finalizing the release scope.

Summary


VALUE
Why build?
For whom?
REQUIREMENT
What world effect
must hold?
ASSUMPTION
Why should the
mechanism
work?
RISK
What if it
does not?
CONTROL +
EVIDENCE
How do we learn
and limit harm?
OWNER +
DECISION
Who may
release,
narrow, or stop?

Prepare for Lecture 3

- Read: Textbook Ch. 8~10
- Prepare a one-page architecture sketch showing
  - the user and the bounded system purpose;
  - major components, data sources, and information flows;
  - the model and its deployment boundary;
  - at least one human decision or oversight point;
  - two important quality attributes;
  - one likely failure and where it should be controlled.